# CSE320: Data Communications

**Quiz 03 [Summer 2025]**

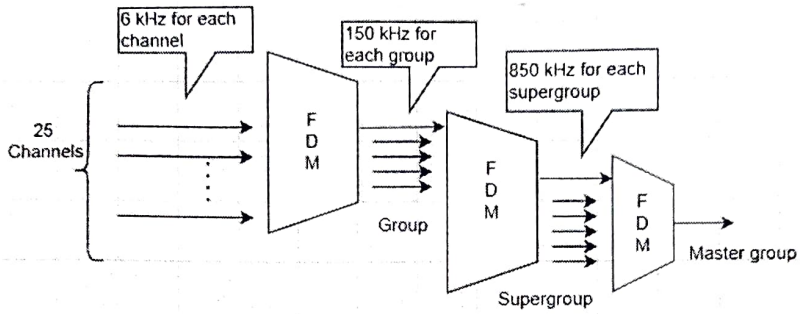
**Time: 20 minutes, Full Marks: 15**

**A**

| Name: | | ID: | | Section: | |
|---|---|---|---|---|---|

1. The following figure shows an analog hierarchical system used by a telephone company for multiplexing. The bandwidth of each link is shown in the box above the links.

**Analog Hierarchy:**




A. Calculate the **total number of channels** in the **master** group. **[2]**

B. Calculate the **guard band** used between the **groups** before forming the supergroup. **[3]**

C. If the guard band between each supergroup is **180 kHz**, calculate **the total bandwidth required** in the master group. **[2]**

A. # channels = $25 \times 5 \times 6 = 750$

B. 5 groups ⟶ 4 guared bands

$\therefore$ Each guared band $= \frac{850 - 5 \times 150}{4}$

$= 25$ kHz

C. Total b/w in master group $= 850 \times 6 + 180 \times 5$

$= 6000$ kHz

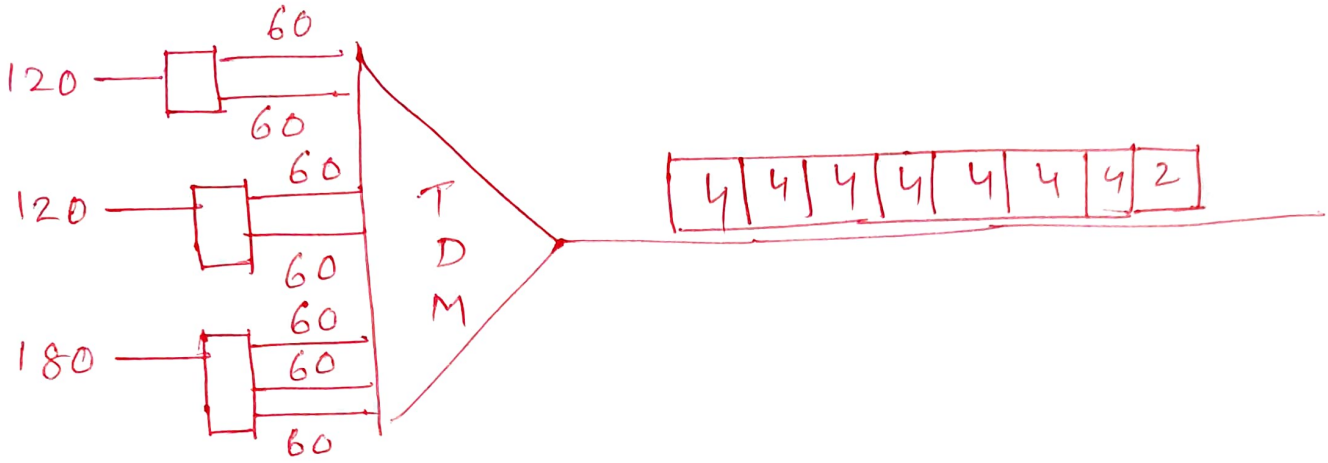
$= 6$ MHz

2. **Three** channels, **ONE** with a bit rate of **180 kbps** and **TWO** with a bit rate of **120 kbps**, are to be multiplexed, with **TWO** synchronization bits and **FOUR** interleaved bits (multiplexing unit). You cannot use pulse stuffing.

A. Show how this can be achieved (figure only). **[3]**

B. What will be the frame size, frame rate & frame duration? **[3]**

C. What will be the output rate and output **slot** duration? **[2]**




B. frame size = $7 \times 4 + 2 = 30$ bits

frame rate = $\frac{60{,}000}{4} = 15{,}000$ fps

frame duration = $\frac{1}{15{,}000} = 6.67 \times 10^{-5}$ s

C. O/P rate = $30 \times 15{,}000 = 450{,}000$ bps

$= 450$ kbps

o/p slot duration = $\frac{\text{frame duration}}{\#\text{ slots}}$

$= \frac{6.67 \times 10^{-5}}{7}$ s

$= 9.53 \times 10^{-6}$ s

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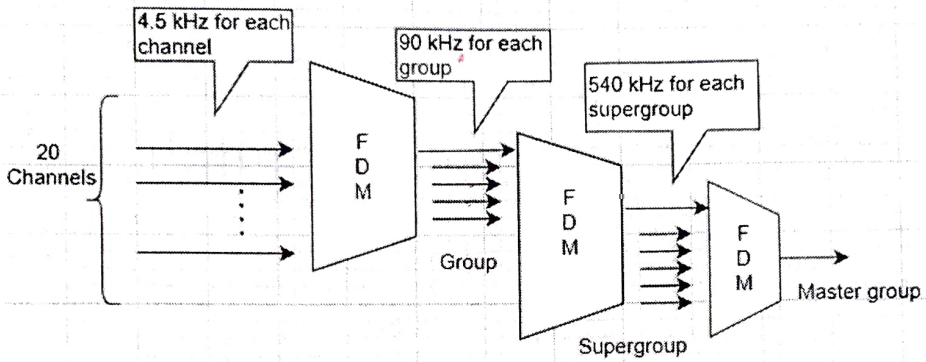
**Time: 20 minutes, Full Marks: 15**

**B**

| Name: | | ID: | | Section: | |
|---|---|---|---|---|---|

1. The following figure shows an analog hierarchical system used by a telephone company for multiplexing. The bandwidth of each link is shown in the box above the links.

**Analog Hierarchy:**




A. Calculate the **total number of channels** in the **master** group. **[2]**

B. Calculate the **guard band** used between the **groups** before forming the supergroup. **[3]**

C. If the guard band between each supergroup is **200 kHz**, calculate **the total bandwidth required** in the master group. **[2]**

A. #channels $= 20 \times 5 \times 6 = 600$

B. g.band $= \frac{540 - 5 \times 90}{4} = 22.5$ kHz

C. b/w in master group $= 540 \times 6 + 200 \times 5$

$= 4240$ kHz

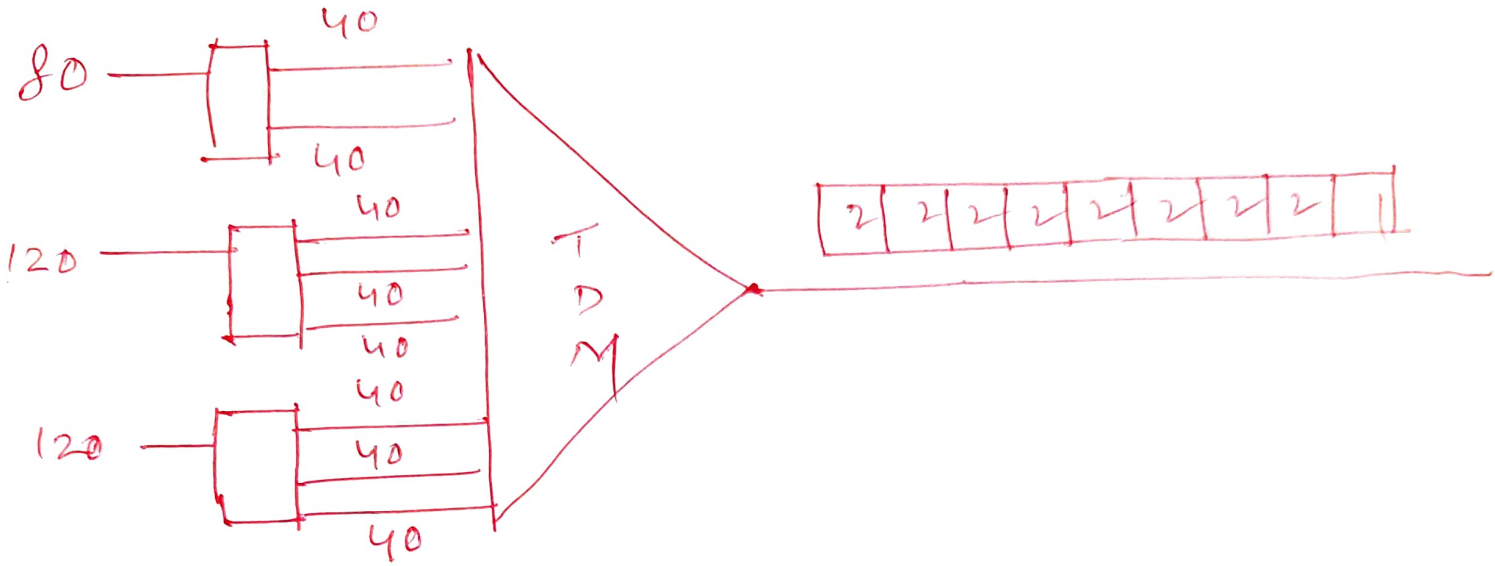
$= 4.24$ MHz

2. **Three** channels, **TWO** with a bit rate of **120 kbps** and **ONE** with a bit rate of **80 kbps**, are to be multiplexed, with **ONE** synchronization bit and **TWO** interleaved bits (multiplexing unit). You cannot use pulse stuffing.

A. Show how this can be achieved (figure only). **[3]**

B. What will be the frame size, frame rate & frame duration? **[3]**

C. What will be the output rate and output **slot** duration? **[2]**




B. f. size = $8 \times 2 + 1 = 17$ bits

f. rate = $\frac{40,000}{2} = 20,000$ fps

f. duration = $\frac{1}{20,000} = 50 \times 10^{-6}$ s

C. o/p rate = $17 \times 20,000 = 340$ kbps

o/p slot dure = $\frac{50 \times 10^{-6}}{8} = 6.25 \times 10^{-6}$ s